

Model Parameter Tables

Table 1: Group 1: well constrained parameters, fixed in all runs.

Symbol	Description	Value	Units	Reference	Notes
ρ_f	Seawater (fluid) density	1.0275	g cm^{-3}	Millero and Poisson [1981]	<i>Standard seawater. Millero & Poisson provide the international equation of state; value changes $< 0.1\%$ over North Atlantic conditions.</i>
ν	Kinematic viscosity	0.01	$\text{cm}^2 \text{s}^{-1}$	Sharqawy et al. [2010]	<i>Seawater at 20 °C. Varies $\sim 30\%$ between 5 and 25 °C; currently held fixed (temperature scaling for future work).</i>
g	Gravitational acceleration	980	cm s^{-2}	—	<i>Standard value; invariant for ocean column depths of interest.</i>
k_B	Boltzmann constant	1.3×10^{-16}	erg K^{-1}	—	<i>NIST fundamental constant; enters only the Brownian kernel.</i>
a_w	Settling speed coefficient (kriest_8)	66	$\text{m day}^{-1} \text{cm}^{-b_w}$	Kriest [2002]	<i>Power-law fit to in-situ Atlantic settling data (Eq. 8 of Kriest 2002). Coefficient and exponent chosen together; treat as a pair.</i>
b_w	Settling speed exponent (kriest_8)	0.62	—	Kriest [2002]	<i>As above. Exponent is robust across datasets; coefficient varies by $\sim$factor 2 between cruises.</i>
$\Delta\rho_{fp}$	Fecal pellet excess density	0.15	g cm^{-3}	Turner [2002]; Ploug et al. [2008]	<i>Dense compact pellets ($\sim 150 \text{ kg m}^{-3}$ above seawater). Gives $\sim 69 \text{ m day}^{-1}$ at bin 8 via Stokes law, consistent with Turner (2002) review and Ploug et al. (2008) measurements.</i>

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Table 1 continued...

Symbol	Description	Value	Units	Reference	Notes
d_0	Monomer (unit particle) diameter	20×10^{-4}	cm	Burd and Jackson [2009]	<i>20 μm monomer, typical phytoplankton cell. Sets the size grid; changes shift all bin boundaries.</i>
A_D	D_{max} amplitude in $D_{\text{max}} = A_D \varepsilon^{-1/4}$	9.39×10^{-6}	$\text{m}^{3/4} \text{W}^{-1/4}$	Burd and Jackson [2009]	<i>From Kolmogorov turbulence theory. D_{max} sets the largest stable aggregate size per depth layer. Amplitude is better constrained than the exponent.</i>

Table 2: Group 2: parameters with reasonable literature estimates but site-dependent values. Held at literature defaults in the first optimization pass; fit in a second pass.

Symbol	Description	Value	Units	Reference	Basis / confidence
D_f	Particle fractal dimension	2.33	—	Logan and Wilkinson [1990]; Li and Logan [1995]	<i>Ocean aggregates: range 1.7–2.5 across studies. Value 2.33 is a mid-range choice consistent with marine snow measurements. Controls mass–size relationship and kernel collision cross-sections.</i>
r/r_g	Ratio of collision radius to radius of gyration	1.6	—	Li and Logan [1997]	<i>Sets effective collision cross-section for shear and differential settling kernels. Li & Logan (1997) derive this from fractal geometry; value varies ~ 1.3–2.0.</i>
Z_c	Filter feeder (macrozooplankton) concentration	0.307	ind m ^{−3}	Stemmann et al. [2004a]; Stemmann et al. [2004b]	<i>Atlantic maximum from Stemmann et al. (2004) Table 2. Depth profile from Fig. 1 wired into <code>DepthProfile.typical()</code>. Needs EXPORTS net tow data (Amy / Debbie) for refinement.</i>
c	Filter feeder clearance rate	0.025	m ³ ind ^{−1} day ^{−1}	Stemmann et al. [2004a]	<i>From Stemmann et al. (2004). Order-of-magnitude known; exact value uncertain.</i>
Z_f	Flux feeder concentration	0.063	ind m ^{−3}	Stemmann et al. [2004a]; Stemmann et al. [2004b]	<i>Atlantic maximum from Stemmann et al. (2004) Table 2. Same depth profile logic as Z_c.</i>
s	Flux feeder capture cross-section	1.3×10^{-5}	m ² ind ^{−1}	Stemmann et al. [2004a]	<i>Stemmann et al. (2004). Interpreted as the projected area presented by a sinking particle to a flux feeder.</i>

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Table 2 continued...

Symbol	Description	Value	Units	Reference	Basis / confidence
p	Egestion fraction (fecal / ingested)	0.3–0.5	—	Stemmann et al. [2004a]	Range reflects food quality and organism type. Default 0.5 (current). Controls fecal pellet production; linked to carbon budget closure.
r_0	Microbial remineralization base rate	0.03	day^{-1}	Iversen and Ploug [2013]	Iversen & Ploug (2013) measured respiration on sinking diatom aggregates in a Norwegian fjord. Value represents warm surface conditions; deep column rates $\sim 10\times$ lower after Q_{10} scaling.
Q_{10}	Temperature sensitivity of microbial rate	2.0	—	Iversen and Ploug [2013]	Standard marine bacteria value. Doubles the rate per 10°C increase. Widely used; uncertainty ± 0.5 .
s_m	Micro-zooplankton mining capture cross-section	1.3×10^{-5}	$\text{m}^2 \text{ind}^{-1}$	Stemmann et al. [2004a]	From Stemmann et al. (2004) Part I Eq. 25; same value as flux feeder cross-section.
f_{next}	Fraction of disaggregated biovolume sent to next smaller bin	2/3	—	Burd and Jackson [2009]	Sectional model redistribution convention. Remainder spread uniformly across smaller bins. No direct measurement; value is a modeling assumption.

Table 3: Group 3: parameters with no reliable direct field estimate. These are fit first against EXPORTS UVP particle size distributions (North Atlantic station, surface to 500 m). Grid search over each parameter individually; conjugate gradient or simulated annealing for joint optimization if needed.

Symbol	Description	Current value	Units	Reference	Why unknown / fit range
α	Collision efficiency (stickiness) of marine snow aggregates	1.0	—	Jackson [1990]; Burd and Jackson [2009]	<i>No direct measurement for natural marine particles. Jackson (1990) set $\alpha = 1$ (upper bound). TEP-coated aggregates may have $\alpha \sim 0.01$–1.0; value depends on surface chemistry and particle type. Fit first: grid search $\alpha \in [0.1, 1.0]$, step 0.1.</i>
α_{cross}	Stickiness between fecal pellets and marine snow	0.5	—	—	<i>No field or lab measurement exists for fecal pellet – marine snow collisions. Fecal pellets are compact ($D_f \approx 2.8$) and low in TEP, so assumed less sticky than aggregates. Current 0.5 is a conservative starting value. Fit after α; range $[0.1, 1.0]$.</i>
δm	Biovolume removed per encounter (micro-zooplankton mining)	10^{-5}	cm^3	Stemmann et al. [2004a]	<i>Proxy for gut volume of small copepods (<i>Oncaea</i> spp.). Stemmann et al. (2004) Part I Eq. 25 gives the functional form but does not constrain δm directly. Range 10^{-6} to 10^{-4} cm^3. Needs EXPORTS gut-content data (Amy / Debbie). Fit with α_{cross}.</i>

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Symbol	Description	Current value	Units	Reference	Why unknown / fit range
Z_m	Micro-zooplankton miner concentration	250	ind m ⁻³	Stemmann et al. [2004a]	Approximate value from Stemmann et al. (2004) Fig. 1. Depth profile $Z_m(z)$ is currently set uniform; needs EXPORTS net tow data to set the actual profile. Uncertainty large ($\times 2-5$).

Optimization note. The loss function is $\mathcal{L} = \sum_{z,k} [\log_{10} N_{\text{model}}(z,k) - \log_{10} N_{\text{obs}}(z,k)]^2$, where k indexes size bin and z indexes depth layer. Log-space weighting ensures that small and large particles contribute equally. Observations: EXPORTS processed UVP particle size distributions, North Atlantic station. See `scripts/optimize/run_alpha_grid_search.m` for the grid-search implementation and `scripts/optimize/loss_size_dist.m` for the loss function.

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